Polynomial AI

Agentic AI, Distributed Systems & Enterprise Architecture

`Confidential Candidate Assessment Pack | PAI/AR1/003`

Overview & Guidelines

Company Overview: Polynomial AI builds enterprise AI systems, retrieval platforms, agentic workflows, and production-grade applications. This assessment evaluates software engineering, architecture design, problem-solving, scalability, security, and communication skills.

Assessment Rules: Expected duration: 48-72 hours. Candidates may use AI tools but must explain design decisions. Production quality code, documentation, and architecture are mandatory.

Technical Expectations:

Python (Any version) | FastAPI | PostgreSQL | Docker | Streamlit/Gradio | Git | Testing Frameworks | Logging | Configuration Management

Project Requirements

Candidates must complete both of the following core projects. Ensure both are production-ready and fully documented.

Project 1: Multi-Agent Workflow Platform

Core Objective: Implement Planner, Retrieval, Execution and Review agents.

Key Features: Support tool calling, orchestration, approvals, retries, memory, observability, execution tracking and conversation management.

Functional Requirements: Authentication, API design, validation, persistence, error handling, logging, monitoring, deployment and documentation.

Project 2: Autonomous Finance Reconciliation Agent

Core Objective: Process orders, DRCs, payments and refunds. Maintain audit logs and provide observability dashboards.

Key Features: Detect duplicates, identify overpayments, reconcile discrepancies, generate corrective actions, and support approval workflows. Explain safeguards and agent decision-making.

Functional Requirements: Role-based access, reporting, administration, auditability, analytics, security controls and production readiness.

Architecture & Non-Functional Requirements

Mandatory Deliverables

- High-Level Architecture Diagram
- Deployment Diagram
- Component Diagram
- Database ER Diagram
- Sequence Diagrams
- API Documentation & README

Non-Functional Needs

- Scalability & Security
- Observability & Monitoring
- Resilience & Fault Tolerance
- Maintainability & Auditability
- Performance Optimization

Submission & Evaluation

Expected Artifacts: Git Repository, README, Demo Video (5-10 mins), Deployment URL (optional), Flowcharts, Sequence Diagrams, Architecture Document, and Trade-off Analysis.

Criteria	Marks	Criteria	Marks
Architecture	20	**Security**	10
Code Quality	15	**Scalability**	10
Database Design	15	**Documentation**	10
AI/LLM Integration	15	**UI / UX**	5

Bonus Challenges: CI/CD, caching, automated testing, monitoring dashboards, advanced analytics, agent workflows, performance optimization, and cloud deployment.